



Rentsyne

Smart Flat Rent & Service Charge Management Platform

Scientific Field: Engineering & Innovation | **Platform:** Android · iOS · Web | **Backend:** Firebase (Cloud Functions, Firestore, FCM)

Project Abstract

Rentsyne is a cloud-based, multi-tenant SaaS platform that digitizes rent and service-charge management for residential buildings in Bangladesh. It connects three user roles — **Master Admin** (platform owner), **Admin** (landlord / property manager), and **Tenant** — through a single Flutter application backed by Firebase. Admins can manage flats, tenants, monthly rent and service charges, and automatically generate PDF invoices, while tenants can view their bills, pay rent online through the SSLCommerz payment gateway, and raise maintenance or service complaints directly from the app. The system replaces manual, paper-based, and spreadsheet-driven rent collection with a transparent, automated, real-time digital workflow.

Objectives

1. Digitize the entire flat rent and service-charge collection process for landlords and tenants.
2. Provide a secure online payment channel (SSLCommerz) so tenants can pay rent without physical visits or cash handling.
3. Automate monthly bill/invoice generation with a clear, itemized summary of dues for every tenant.
4. Create a centralized complaint / service-request system so maintenance issues are tracked and resolved faster.
5. Give property owners a real-time dashboard to track occupancy, payment status, and tenant records.
6. Enable a Master Admin layer to oversee multiple property admins on one platform, with audit logs, broadcasts, and reporting.
7. Support both Bengali and English so the app is accessible to local users.

Innovation / Novelty

Unlike the spreadsheet- and register-based methods still widely used by small and mid-sized landlords in Bangladesh, Rentsyne is purpose-built for the local rental market:

- A three-tier role system (Master Admin → Admin → Tenant) that enables a scalable, multi-property SaaS model rather than a single-building tool.
- Integrated local payment gateway (SSLCommerz) support, tailored for Bangladeshi banking and mobile financial services.
- Automatic PDF invoice generation and real-time rent-status tracking, eliminating manual bill preparation.
- Tenant-side self-service features (bill history, online payment, complaint submission) that are rare in existing local rent-management solutions.
- Built-in audit logging, maintenance-mode control, and platform-wide broadcast/announcement tools for governance at scale.

Expected Impact

Rentsyne aims to reduce the administrative burden and payment friction currently faced by landlords and tenants in Bangladesh’s rental housing sector. By digitizing rent collection, it can reduce late or missed payments, eliminate manual bookkeeping errors, and provide transparent, verifiable payment records for both parties. For tenants, it improves convenience and trust through instant bill visibility and online payment; for landlords, it saves time, reduces disputes, and provides data-driven insight into their properties. At scale, the platform can support the modernization of property management across urban and semi-urban housing markets in Bangladesh, with potential to expand into broader property/facility management services.

Key Features

For Admins (Landlords)

- Flat & tenant management
- Automated rent & invoice generation
- Payment status tracking
- Maintenance / complaint handling
- Reports & data export

For Tenants

- Monthly bill summary
- Online rent payment (SSLCommerz)
- Service request submission
- Notifications & announcements
- Bengali / English language support

For Master Admin

- Multi-admin platform oversight
- Audit log & activity tracking
- Platform-wide broadcasts
- Feedback & growth reporting

Technology Stack

- Flutter (Android, iOS, Web)
- Firebase (Auth, Firestore, Functions, FCM, App Check)
- SSLCommerz payment gateway
- PDF invoice generation